



DEPARTMENT OF CHEMICAL ENGINEERING NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

BIORESOURCE TECHNOLOGY (CET-417)

Home Assignment plus Question Bank-3

S. No.	Questions	COs
1.	Explain gasification and the chemical reactions along with various steps involved in the process.	CO2
2.	What are gasifiers? Discuss updraft, downdraft and fluidized bed gasifiers with the help of schematic diagrams.	CO3
3.	Define equivalence ratio and energy efficiency of gasifier and explain their significance.	CO2
4.	Discuss typical zonal chemical reactions (Mechanisms) and products in a co-current and counter current gasifiers.	CO3
5.	Elaborate advantages and disadvantages of gasification along with the desirable properties of biomass (factors) effecting gasification.	CO2
6.	Describe some important design considerations for the fixed bed stationary gasifiers.	CO3
7.	What is pyrolysis? Discuss the process with the help of a simple schematic diagram.	CO2
8.	What are various possible feedstocks for pyrolysis? Explain the mechanism by drawing a neat sketch.	CO2
9.	Discuss different pyrolysis products along with their applications.	CO2
10.	Write a brief note on syngas.	CO2
11.	Enlist various types of pyrolysis, and explain fixed-bed and fluidized bed systems by sketching figures and compare them with respect to their overall performance and efficiency.	CO3
12.	What are the current status and future challenges for pyrolysis technology?	CO2
13.	Explain anaerobic digestion and discuss the pathways of degradation of the organic materials with the help of a diagram.	CO2
14.	Briefly elaborate various factors effecting anaerobic digestion.	CO2
15.	What are the products of anaerobic digestion? Discuss the advantages and disadvantages of anaerobic digestion and co-digestion.	CO2
16.	What is biogasification? Discuss the Indian scenario with respect to cow dung.	CO2
17.	Sketch the diagram of an Indian floating dome type gobar gas plant and discuss it.	CO3
18.	Enlist and explain different factors effecting the biogasification of cow dung.	CO2
19.	Write a short note on the biogas and discuss the recent development in biogasification.	CO2
20.	Suppose you have a dairy farm which houses 50 grown up cows and 40 grown up buffalos. Whereas the cows and buffaloes have the same number of calves. In order to produce biogas from the animal dung, give the design of a gobar gas plant with the help of the following data: (a) Average gobar available/cow/day = 10 kg; (b) Average gobar available/buffaloe/day = 15 kg (c) Average biogas available/kg gobar/day = 0.035 m ³ (d) Average percentage of solid in gobar = 20 (e) Average bulk density of gobar = $P_G = 1200 \text{ kg/m}^3$ (f) The maximum allowable capacity of the digester for a small plant = 60 m ³ Optimum ratio of water to gobar in the slurry = 1:1	CO4
CO1:	Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability).	
CO2:	Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.	
CO3:	Exhibiting knowledge of the systems used for bioresources and bioresource technology.	
CO4:	Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.	